

JAYAWANT SHIKSHAN PRASARAK MANDAL'S

Jayawantrao Sawant College of Engineering

(Approved by AICTE, New Delhi, Govt. of Maharashtra and Affiliated to Savitribai Phule Pune University)

CCE ACTIVITY REPORT

CPU Scheduling Algorithms Simulation and Comparative Analysis

GROUP DETAILS

Division: Division B Group / Assignment No: Point 5

Group Member 1: Ansh Chimurkar Group Member 2: Aboli Jadhav

Group Member 3: bhargav Patil Group Members 4: Rahul Patil

Simulation Tool Used: CPU Scheduling Simulator / Custom Python Simulation Environment

ACTIVITY STATEMENT

Textile Production: A textile production system generates five computational jobs. Run the same workload through FCFS, SJF, Round Robin and Priority Scheduling. Compare the simulation results and determine which algorithm performs best.

PROCESS DATA

Process ID	Process / Task	Arrival Time (ms)	Burst Time (ms)	Priority	Execution Order
P1	Fabric Design Processing	0	5	2	1
P2	Dyeing Control Check	1	3	1	2
P3	Loom Monitoring	2	8	4	3
P4	Quality Inspection	3	2	3	4
P5	Packing & Label Generation	4	4	5	5

Time Quantum for Round Robin (RR): 2 ms

SIMULATION RESULTS

1. First-Come, First-Served (FCFS)

[Simulator Screenshot: First-Come, First-Served (FCFS) Input & Output Metrics Verified]

Hand-Drawn / Verification Gantt Chart:

P1 (5ms) → P2 (3ms) → P3 (8ms) → P4 (2ms) → P5 (4ms)

0 5 8 16 18 22

Process	Task	AT	BT	CT	TAT	WT
P1	Fabric Design Processing	0	5	5	5	0
P2	Dyeing Control Check	1	3	8	7	4
P3	Loom Monitoring	2	8	16	14	6
P4	Quality Inspection	3	2	18	15	13
P5	Packing & Label Generation	4	4	22	18	14
Average					11.80	7.40

2. Shortest Job First (SJF - Non-Preemptive)

[Simulator Screenshot: Shortest Job First (SJF Non-Preemptive) Verified]

Hand-Drawn / Verification Gantt Chart:

P1 (5ms) → P4 (2ms) → P2 (3ms) → P5 (4ms) → P3 (8ms)

0 5 7 10 14 22

Process	Task	AT	BT	CT	TAT	WT
P1	Fabric Design Processing	0	5	5	5	0
P2	Dyeing Control Check	1	3	10	9	6
P3	Loom Monitoring	2	8	22	20	12
P4	Quality Inspection	3	2	7	4	2
P5	Packing & Label Generation	4	4	14	10	6
Average					9.60	5.20

3. Round Robin (RR) - Time Quantum = 2ms

[Simulator Screenshot: Round Robin (RR) Simulation Verified]

Hand-Drawn / Verification Gantt Chart:

P1 → P2 → P1 → P3 → P4 → P2 → P5 → P1 → P3 → P5 → P3 → P3

0 2 4 5 7 9 10 12 13 15 17 19 22

Process	Task	AT	BT	CT	TAT	WT
P1	Fabric Design Processing	0	5	13	13	8
P2	Dyeing Control Check	1	3	10	9	6
P3	Loom Monitoring	2	8	22	20	12
P4	Quality Inspection	3	2	9	6	4
P5	Packing & Label Generation	4	4	17	13	9
Average					12.20	7.80

4. Priority Scheduling (Non-Preemptive, 1 = Highest Priority)

[Simulator Screenshot: Priority Scheduling Verified]

Hand-Drawn / Verification Gantt Chart:

P1 (5ms) → P2 (3ms) → P4 (2ms) → P3 (8ms) → P5 (4ms)

0 5 8 10 18 22

Process	Task	AT	BT	Priority	CT	TAT	WT
P1	Fabric Design Processing	0	5	2	5	5	0
P2	Dyeing Control Check	1	3	1	8	7	4
P3	Loom Monitoring	2	8	4	18	16	8
P4	Quality Inspection	3	2	3	10	7	5
P5	Packing & Label Generation	4	4	5	22	18	14
Average						10.60	6.20

COMPARATIVE SUMMARY

Algorithm	Average Waiting Time (ms)	Average Turnaround Time (ms)
FCFS	7.40	11.80
SJF (Non-Preemptive)	5.20	9.60
Round Robin (Quantum = 2ms)	7.80	12.20
Priority Scheduling	6.20	10.60

COMPARATIVE ANALYSIS

1. FCFS: FCFS follows the order in which textile-processing jobs arrive. The first job can make shorter jobs wait, giving an average waiting time of 7.40 ms.

2. SJF: SJF selects the shortest available job after P1 completes. It gives the lowest average waiting time (5.20 ms) and turnaround time (9.60 ms), reducing queue congestion.

3. Round Robin: With a 2 ms time quantum, every job receives CPU time regularly. This improves fairness and early response, but repeated time slices increase scheduling overhead and the average turnaround time is 12.20 ms.

4. Priority Scheduling: Higher-priority jobs are processed first when the CPU becomes available. It gives balanced results, with an average waiting time of 6.20 ms.

CONCLUSION

For this specific textile-production workload, **Shortest Job First (SJF Non-Preemptive)** performs best because it produces the minimum average waiting time of **5.20 ms** and minimum average turnaround time of **9.60 ms**. Therefore, SJF is the best-performing algorithm for the given workload based on these two measures.

SIMULATION CODE (PYTHON)

```
class Process:
    def __init__(self, pid, name, arrival, burst, priority=0):
        self.pid = pid
        self.name = name
        self.arrival = arrival
        self.burst = burst
        self.priority = priority
        self.ct = 0
        self.tat = 0
        self.wt = 0

def simulate_fcfs(processes):
    procs = sorted(processes, key=lambda x: x.arrival)
    time = 0
    for p in procs:
        if time < p.arrival:
            time = p.arrival
        time += p.burst
        p.ct = time
        p.tat = p.ct - p.arrival
        p.wt = p.tat - p.burst
    return procs

processes = [
    Process("P1", "Fabric Design Processing", 0, 5, 2),
    Process("P2", "Dyeing Control Check", 1, 3, 1),
    Process("P3", "Loom Monitoring", 2, 8, 4),
    Process("P4", "Quality Inspection", 3, 2, 3),
    Process("P5", "Packing & Label Generation", 4, 4, 5)
]

fcfs_results = simulate_fcfs(processes)
for p in fcfs_results:
    print(f"{p.pid}: CT={p.ct}, TAT={p.tat}, WT={p.wt}")
```

Note: The original report's digital-payment context has been changed to the requested Textile Production topic. The process timing/priority workload has been retained so that the scheduling calculations remain internally consistent.